

RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN
HIGHER NATIONAL DIPLOMA EXAM (HND)

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION

NATIONAL COMMISSION FOR THE ORGANISATION OF HIGHER
NATIONAL DIPLOMA EXAM (HND) EXAM

National Exam of Higher National Diploma-New program – 2021 Session

Spécialité/option : Software Engineering (SWE)

Paper : System Analysis and Design

Duration : 4Hours

Credits : 7

Instruction: Answer all questions

SECTION A: INFORMATION SYSTEM AND DATA BASE (50 marks)

Part I: Object modeling (25 marks)

1. What is a model? (1mark)
2. Why is it important to always model computer system? (2marks)
3. What is Object Oriented Modelling? (2marks)
4. According to you what is UML Language? (2marks)
5. To understand UML diagrams and to learn UML, the user must have knowledge about the conceptual model of UML. The conceptual model consists of three parts.
Name them (3marks)
6. UML Architecture can be best represented as a collection five views. Name the five (05)
(3marks)
7. What is association? Illustrate with an example the concept of association. (2marks)
8. What is multiplicity in associations? Give example to explain multiplicity. (2marks)
9. Create the use case diagram according to the following elevator control system functional requirements: (2X4=8 marks)
 - a. The elevator control system shall allow the passenger to call the elevator and to select the destination floor;
 - b. When the passenger pushes the external button (to call the elevator), or the internal button (to select the destination floor), the central control system switches the button light on;
 - c. When the passenger calls the elevator or selects the destination floor, the central control system opens/closes the elevator door;
 - d. When the passenger calls the elevator or selects the destination floor, the central control system moves/stops the elevator to/at the passenger call floor or to/at the passenger destination floor.

When the passenger leaves the elevator, the central control system switches the button light off

SECTION B: DATABASE: (25MARKS)

1. Based on the given table, solve the following queries.

Employee table

empId	empName	Age	Address
1001	Mekole	26	Delhi
1002	Etamo	30	Texas
1003	Ankit	27	Mumbai
1004	Gad	32	Nagpur

- Write the SELECT command to display the details of the employee with empId as 1004. (1mark)
 - Write the SELECT command to display all the records of table Employees. (2marks)
 - Write the SELECT command to display all the records of the employee whose name starts with the character 'R'. (2marks)
2. Consider the ER model given in Figure 1. This model represents the operations of a pharmacy chain. Please answer the following questions regarding this model.
- Can a pharmaceutical company have multiple phone numbers? If not, what do you need to do to allow this? (5marks)

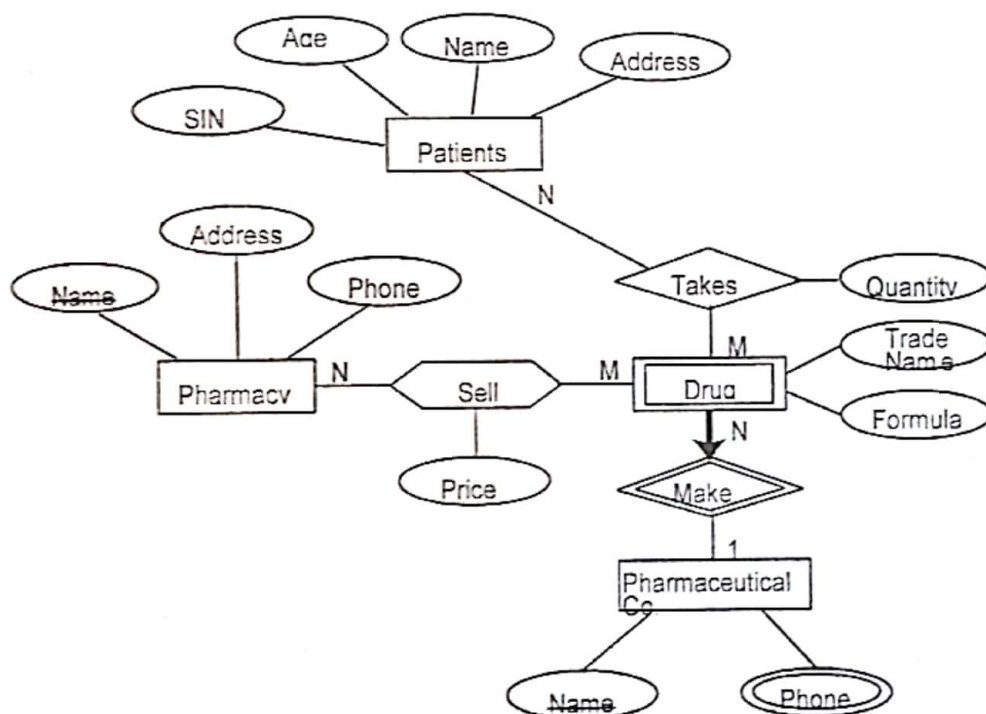


Figure 1

- Modify the model (by adding to Figure 1, *not* by drawing another figure) so that you can represent the following

- Each patient has to have one and only one primary physician. Each physician has at least one patient. We want to know at least the specialty and the date of entry into the profession of each physician.
 - Instead of modeling only the fact that a patient takes certain drugs, model the fact that a patient takes certain drugs that are prescribed by a physician and the prescription date.
 - Pharmaceutical companies have long-term contracts with pharmacies. A pharmaceutical company can contract with several pharmacies, and a pharmacy can contract with several pharmaceutical companies. For each contract we want to store a start date, an end date. (5marks)
3. What is Data Integrity in database? (2marks)
 4. What are Entities and Relationships? (3marks)
 5. What is an Alias in SQL? (2marks)
 6. What are the various forms of Normalization? (3marks)

Part II: WEB PROGRAMMING (25 MARKS)

1. What are tags? Give two HTML tags that don't come in a pair. (4 marks)
2. What are style sheets? Explain three ways of applying style sheets in a web document. (5marks)
3. What difference do you make between the followings: (4 marks)
 - Intranet and extranet?
 - Web page and web document?
4. Define Internet and state four of its services. (4 marks)
5. What are some of the common lists that can be used when designing a page? (5 marks)
6. Write an HTML table tag sequence that outputs the following: (3 marks)

50 pcs	100	500
10 pcs	5	50

II-MOBILE PROGRAMMING (25 marks)

1. What are the main languages supported by Android for application development? (2marks)
2. Do all mobile phones support the latest Android operating system? Justify (2marks)
3. What is the difference between Mobile device testing and mobile application testing? (2marks)
4. What are the different states wherein a process is based? (4marks)
5. What are the four essential states of an activity? (4marks)
6. Android Architecture is made up of 4 key components. List them (4marks)
7. What do you understand by Native Mobile Application Development? (2marks)
8. Describe Android application Architecture? (5 marks)